SETUP

INSTALL GUIDE

Install the device, run the first-load check, and rebuild after you edit the engine.

Dub Assistant is a Max for Live MIDI effect that maps three semantic knobs onto Live's mixer the way a dub engineer uses a console: one gesture moves many sends and faders. You train it on **your** mix moves. It is an assistant, not an autonomous mixer.

The interaction model follows Joaquín Jiménez-Sauma, *Creating a Machine Learning Assistant for the Real-Time Performance of Dub Music* (CSMC / MuMe 2020). The 2012 ML.* library is replaced by a small built-in multilayer perceptron so the device trains on this PC without extra Max packages.

This folder is the project. The playable device is also copied into your Ableton User Library.

> IMPORTANT
>
> Put the device on a dedicated MIDI track named **Dub Assistant**. Do not put it in an audio path. It talks to the mixer through the Live Object Model. It does not process sound.

WHAT YOU GET

The install puts these files in two places: this project folder, and A:\User Library\Presets\MIDI Effects\Max MIDI Effect\Dub Assistant\.

File	Role
Dub Assistant.amxd	Max MIDI Effect you load in Live
dub-assistant.js	Engine: dataset, MLP, mixer writes, fader motion
Dub Assistant.maxpat	Same patch, for editing in Max
sample-dataset.csv	30 starter rows so Train works on first launch
docs/USER-GUIDE.md	How to train, perform, and map the Launchkey
docs/OPERATOR-CARD.md	One-page cheat sheet at the desk
template/TEMPLATE-SET.md	How to build the 8-track + 2-return Set

REQUIREMENTS

You need the following on this Windows machine:

- Ableton Live 12 with Max for Live (this studio: Live 12.4.3)
- The User Library at A:\User Library (already set in Live's Library.cfg)
- Eight audio tracks named **Drums**, **Bass**, **Vox**, **Keys**, **Guitar**, **FX**, **Perc**, **Other** (or rename them in the device)
- At least one return track. The device writes send **A** by default (Delay)

Optional:

- Live 12 Suite, if you want Stem Separation (chapter 22). This device cannot trigger the split. You run it, then load the files.
- Novation Launchkey MK4 61 for the five performance knobs (chapter 33)
- Producer Pal, if you want the template Set built for you

You do **not** need FluCoMa or the old ML.* package for v1. Training runs inside dub-assistant.js.

INSTALL

The device files are already in your User Library if this project was built on this PC. Confirm, then do a first-load check.

1. Confirm this folder exists and contains Dub Assistant.amxd plus dub-assistant.js: A:\User Library\Presets\MIDI Effects\Max MIDI Effect\Dub Assistant\
2. If that folder is missing, copy the four files from device\ in this project into that User Library path. Create the folders if Live has never saved a Max MIDI Effect there.
3. Keep dub-assistant.js in the **same folder** as Dub Assistant.amxd. Max loads the script from beside the device.
4. Restart Live, or click the Browser's refresh control, so User Library re-indexes.
5. In Live's Browser, open **User Library → Presets → MIDI Effects → Max MIDI Effect → Dub Assistant**.
6. Drop **Dub Assistant** onto a new MIDI track. Name that track Dub Assistant.
7. Read the status line at the top of the device. After a moment it must say ready or bound N/8 tracks.

> NOTE
> If the status line stays blank, open the Max window from the device's title bar and check the Max console for dub-assistant: lines. The usual cause is a missing dub-assistant.js next to the .amxd.

FIRST-LOAD CHECK

Run this once after install so you know the mixer writes work before you train anything.

1. Build or open a Set with the eight named audio tracks and a return named **Dub Delay**. See **Template Set**.
2. On the device, click **Bind**. Status must report how many of the eight names it found.
3. Turn **Dry/Wet** to 100%.
4. Turn **col**. Send A on the bound tracks must move.
5. Start Live's transport. Faders must start making occasional non-repeating moves on 1/8 notes. Turn **Dry/Wet** back to 0%. The mixer must stop moving.
6. Click **Load**, choose sample-dataset.csv, then click **Train**. Status must say trained on 30 rows.

If step 4 does nothing, the bind list is empty. Rename the audio tracks to match the labels, or send names from Max (see the user guide).

REBUILD THE DEVICE

If you edit scripts/build_device.py or device/dub-assistant.js, rebuild and recopy:

```
python "B:\My Apps\Windows apps by me\Dub-Assistant\scripts\build_device.py"
```

Then reload the device in Live (delete it from the track and add it again, or toggle it off and on).

WHAT THIS DEVICE DOES NOT DO

The following are out of scope for v1, on purpose:

- Stem Separation (you trigger that in Live; Pal cannot either)
- VST or AU parameters
- Arrangement automation or clip envelopes
- Audio-feature inputs (the paper lists that as further work)
- Recreating the Polymer Dub composition. This is the control device only

NEXT STEPS

Read **User Guide** before the first real training session. Keep **Operator Card** open on the desk while you play.